MILITARY SPECIFICATION

INERT FILLERS FOR USE IN AMMUNITION

This Amendment forms a part of Military Specification MIL-I-60350 (MU), dated 30 October 1965, and is approved for use by the US Army Armament Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

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* 1.2, Classification: Add the following types of inert fillers:

"Type IV - Pelron (polyurethane) inert filler (see 6.2, 6.13 and 6.14).

Type V - A mixture of sand and steel punchings (see 6.2 and 6.15)."

* 2.1 SPECIFICATIONS: Add the following Military Specifications:

"MILITARY

MIL-I-706	- Iron Oxide: Ferric, Red Dry
MIL-G-20098	- Gypsum, Calcined.
MIL-B-48177	- Barium Sulfate.
MIL-C-48244	- Cellulose, Powdered
MIL-G-48246	- Glyceryl Ester of 12 - Hydroxy Stearic Acid.
MIL-G-48334	- Gypsum, Dead Burned

* Delete the following:

"PURCHASE DESCRIPTIONS (See 6.3)
PA-PD-290 - Gypsum, Calcined"

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* Delete the following:

"PA-PD-656 - Iron Oxide, Ferric
PA-PD-721 - Glyceride of Hydroxy Stearic acid
PA-PD-722 - Gypsum, dead burned
PA-PD-2765 - Cellulose, powdered
PA-PD-2766 - Barium Sulfate"

3.1.2, Types:

Type II, Ferric Oxide, Add: "or Type I, Class 2"

Add the following:

"Constituent Material	Conforming to	Parts by Weight
Prepolymer fraction	Pelron Technical Bulletin 2012	21
Resin - Catalyst fraction	Pelron Technical Bulletin 2012	79

Type V

Constituent Material	Conforming to	Parts by Weight
Sand	(See 6.15)	50
Steel punchings	(See 6.15)	50 "

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6.2, Add new subparagraphs as follows:

"6.2.4, Type IV Filler - Type IV filler is suitable for a all applications.

6.2.5 Type V Filler shall be retained as a secondary inert filler for use at US Army proving grounds with limited application such as weapons proof testing and in other cases where interior ballistics (exclusive of maximum chamber pressures) and exterior ballistics are not of major concern in test firings."

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* 6.3, PA-PD documents; Delete paragraph 6.3 in its entirety.

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6.11, third sentence, change "Report No. 3276" to "Report No. 3283".

Add new paragraph:

"6.13 Type IV filler - Type IV (Pelron Polyurethane) filler is formed by combining two materials (prepolymer and resin-catalyst fractions) at the mixing head of a dispensing machine. Within approximately one minute, chemical reaction causes the material to expand. Projectiles are placed on a scale under the dispensing nozzle and are filled until the total weight of the projectile and filler is correct. The material fills the projectile cavity to approximately 60% of its capacity. The inert fuze or projectile nose plug, or cavity mold, is then screwed securely in place. Chemical reaction expands the material to completely fill the cavity with a homogenous material that ensures correct density and center of gravity (see US Army Jefferson Proving Ground, Report No. MTD-1-5-75, June 1975). The prepolymer and resin-catalyst fractions are produced by Pelron Corporation, located at 7847-W47 St. Lyons, IL 60534."

Add new paragraph:

"6.14 CAUTION NOTICE - During the formation of Pelron (polyurethane) filler, fumes containing toluene diisocyanate, which is considered toxic, are present at the dispensing nozzle. In addition, methylene chloride is used as a cleaning agent. Therefore, during formation and loading of the Pelron filler, adequate ventilation must be provided and breathing of the vapor must be avoided. A hood type ventilation shall be situated over equipment where the vapors are exposed to the atmosphere. The ventilation system should maintain an air velocity of at least 100 ft./min. past the worker. If air sampling discloses that the threshold limit value of toluene diisocyanate (.02 ppm) is exceeded during the operation and further ventilation is not feasible, then employees must wear respiratory protective devices. Care must be taken to avoid contact of the solvents with the eyes or skin. In case of contact with methylene chloride, immediately flush the affected area with plenty of water for at least 15 minutes. If the skin becomes contaminated with toluene diisocyanate, it should

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be immediately washed off with a solution of 5% aqueous ammonia mixed wih 10% isopropyl alcohol and then reported to a physician. In addition to these precautions, the provisions of AMCP385-112 for the "Hazard Analysis of Polyurethane Foams" shall be adhered to."

Add new paragraph:

"6.15 Type V filler - Type V (sand and steel punching) inert filler consists of sand described as "Sand, Blast No. 12 Berkley, CRY, 18 Mesh, Grade No. 1, New Jersey, Silica Glass Type Sand or equal, No. 5350-DO-X04-7671" and steel punching 1/16" to 1/4" thick with a 1" diameter. The steel punching are covered with a slight coat of oil. A source of supply of the steel punching is "Coatesville Plate Washer Company, PO Box 127, Coatesville, Pennsylvania 19320".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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